Topological Selection in Nonlinear Field Theories:

Robust Dominance of Kink Configurations under Structural Energy Weighting

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March 15, 2026

Abstract

We present a numerical study of structural selection in a one-dimensional nonlinear scalar field theory with ϕ^4 potential. Building on the two-level framework introduced in a companion paper [1] — where fundamental dynamics defines the space of possible field configurations and a structural energy functional $\mathcal{E}[\Phi]$ selects physically preferred ones — we ask: which topological class of configurations is structurally preferred?

Four classes of initial configurations are evolved under conservative Klein–Gordon dynamics: homogeneous vacuum states, topological kink solutions, localised wave packets, and chaotic initial data. For each trajectory, the structural energy is evaluated using a lattice-adapted subset of the structural diagnostics, probing stability, spatial connectivity, and dynamical activity. A Boltzmann-type probability weight $P[\Phi] \propto e^{-\beta\mathcal{E}[\Phi]}$ is then assigned to each configuration class.

The central numerical result is robust across all parameter ranges tested: *kink configurations dominate the structural weight across all 81 combinations of Lagrange multiplier and inverse temperature values tested.* Vacuum states, localised packets, and chaotic configurations are consistently disfavoured. This result is consistent with the central prediction of the structural selection framework: topologically stable, moderately active, well-connected configurations are structurally preferred over trivial, dispersive, or chaotically unstable ones.

The study is presented as a proof of concept for the discriminative power of the structural selection principle in a setting where the solution space contains multiple qualitatively distinct classes.

1 Introduction

A central question in theoretical physics is why, among the many mathematically valid solutions of fundamental field equations, only a small subset appears to be physically realised. The companion paper [1] proposes a *structural selection principle*: the dynamics $\delta S[\Phi] = 0$ defines the space of possible configurations, while a covariant structural energy functional $\mathcal{E}[\Phi]$ selects physically preferred ones via a Maximum-Entropy weighting on the solution manifold.

The companion framework introduces five structural diagnostics in full generality [1]. In the present lattice implementation, we use a reduced subset probing three aspects most accessible in the lattice setting:

- $\mathcal{H}$: Residual Stability — distance from the solution manifold
- $\mathcal{N}$: Connectivity — spatial coherence via mutual information
- $\mathcal{S}_C$: Dynamical Activity — regulated non-equilibrium activity

The present paper tests this framework numerically in a concrete setting: a one-dimensional ϕ^4 scalar field theory, which admits four qualitatively distinct classes of solutions: homogeneous vacuum states, topological kink solutions, localised wave packets, and chaotic configurations.

This setting is motivated by observations in complex systems theory [2, 4], where topological defects such as solitons, vortices, and domain walls are known to be among the most persistent field configurations in nonlinear systems. The question arises naturally: does the structural selection framework correctly identify topological stability as a preferred structural property?

The central question is: within the tested benchmark setup, does the structural energy functional $\mathcal{E}[\Phi]$ correctly identify kink configurations as preferred, consistent with their known topological stability and physical persistence?

2 Framework Recap

2.1 Two-Level Architecture

Following [1], the structural selection principle operates in two layers:

Level 1 — Dynamics: $\delta S[\Phi] = 0$ defines the solution manifold $\mathcal{M}_S$.

Level 2 — Selection: A structural energy functional $\mathcal{E}[\Phi]$ induces a Boltzmann-type probability measure on $\mathcal{M}_S$:

$$P[\Phi] \propto \exp(-\beta\, \mathcal{E}[\Phi])\,, \tag{1}$$

where $\beta > 0$ controls the sharpness of the selection. Configurations with lower structural energy are exponentially preferred.

2.2 The Structural Energy Functional

The structural energy is constructed from three lattice-adapted diagnostic sectors relevant for the lattice setting:

$$\mathcal{E}_{\text{lat}}[\Phi] = \lambda_{\text{stab}}\, \mathcal{E}_{\text{stab}} + \lambda_{\text{conn}}\, \mathcal{E}_{\text{conn}} + \lambda_{\text{dyn}}\, \mathcal{E}_{\text{dyn}}, \tag{2}$$

where each sector is defined in Section 3. The weights $\lambda_i > 0$ are treated as free parameters; the robustness of the selection result across a wide range of λ_i values is the central claim of this paper. In brief: low structural energy corresponds to configurations that are simultaneously stable, spatially coherent, moderately active, and structurally consistent.

3 Model and Diagnostics

3.1 Dynamical Model

The field $\phi(x,t)$ evolves on a periodic lattice of $N = 128$ sites with spacing $\Delta x = 1$ under the Klein–Gordon equation with ϕ^4 potential:

$$\ddot{\phi}_i = \frac{\phi_{i+1} + \phi_{i-1} - 2\phi_i}{\Delta x^2} - V'(\phi_i), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2. \tag{3}$$

The dynamics is integrated with a fourth-order Runge–Kutta scheme ($\Delta t = 0.02$, $T_{\max} = 24$). The potential V has two degenerate minima at $\phi = \pm 1$, supporting topological kink solutions $\phi(x) \approx \tanh((x - x_0)/w)$ that interpolate between the vacua.

3.2 Four Configuration Classes

We study four qualitatively distinct initial conditions:

Vacuum: $\phi \approx \pm 1$ with small noise. Homogeneous ground state; no spatial structure.

Kink: $\phi(x) = \tanh((x - x_0)/w)$ with $w \in [4, 10]$, $x_0 \in [-10, 10]$. Topological soliton interpolating between the two vacua.

Localised: Gaussian wave packet with amplitude $A \in [0.8, 1.8]$ and width $\sigma \in [3, 8]$. Non-topological, dispersive configuration.

Chaotic: $\phi, \dot{\phi} \sim \mathcal{N}(0, 1)$. Random initial data with no coherent structure.

For each class, $n = 12$ independent runs are generated from randomised initial conditions.

3.3 Lattice Structural Diagnostics

Stability sector $\mathcal{E}_{\text{stab}}$:

$$\mathcal{E}_{\text{stab}} = \frac{\sum_i \pi_i^2}{\Lambda_{\text{stab}} + \sum_i \pi_i^2} + \frac{\sum_i \max(0, |\phi_i| - \phi_{\max})^2}{\Lambda_{\text{bound}} + \sum_i \max(0, |\phi_i| - \phi_{\max})^2}, \tag{4}$$

with $\Lambda_{\text{stab}} = \Lambda_{\text{bound}} = 10$ and $\phi_{\max} = 3$. Both terms are bounded in $[0, 1)$. Configurations with large kinetic energy or field values outside the physical range are penalised.

Connectivity sector $\mathcal{E}_{\text{conn}}$:

$$\mathcal{E}_{\text{conn}} = -I(\phi_i : \phi_{i+1}), \tag{5}$$

where $I(\phi_i : \phi_{i+1})$ is the mutual information between adjacent field values, estimated from their empirical joint distribution with 16 bins. High mutual information (strong spatial coherence) lowers the structural energy.

Dynamical activity sector $\mathcal{E}_{\text{dyn}}$:

$$\mathcal{E}_{\text{dyn}} = \left(\frac{s_{\text{dyn}} - s_\star}{\sigma_s} \right)^2, \qquad s_{\text{dyn}} = \frac{\sum_i \pi_i^2 + \alpha \sum_i (\nabla\phi)_i^2}{10 + \sum_i \pi_i^2 + \alpha \sum_i (\nabla\phi)_i^2}, \tag{6}$$

with target $s_\star = 0.42$, width $\sigma_s = 0.18$, and $\alpha = 0.5$. The target activity $s_\star$ is a benchmark parameter intended to encode the qualitative preference for regulated non-equilibrium dynamics; it is not derived from first principles, and a sensitivity analysis over $s_\star$ is deferred to future work. This implements a *preferred activity window*: both static ($s_{\text{dyn}} \approx 0$) and chaotically active ($s_{\text{dyn}} \approx 1$) configurations are penalised relative to configurations with moderate, regulated dynamical activity.

The preferred-window formulation is motivated by Prigogine's observation [2] that persistent complex structures require regulated non-equilibrium activity: neither frozen nor chaotic, but operating in a regime of ordered dissipative dynamics.

4 Results

4.1 Class-Level Structural Energies

Table 1 reports the mean structural energy $\bar{\mathcal{E}}$, its standard deviation, the Boltzmann weight sum, and the mean dynamical activity for each class at the reference parameter setting $\lambda_{\text{stab}} = \lambda_{\text{conn}} = \lambda_{\text{dyn}} = 1$, $\beta = 4$.

Table 1: Class-level structural energy summary ($\lambda_i = 1$, $\beta = 4$, $n = 12$ runs per class).

Class	$\bar{\mathcal{E}}$	$\sigma_{\mathcal{E}}$	$\sum P_w$	$\bar{s}_{\text{dyn}}$
Vacuum	4.479	0.050	≈ 0	0.013
Kink	0.015	0.266	0.9996	0.255
Localised	1.941	0.309	4×10^{-4}	0.670
Chaotic	9.210	0.188	≈ 0	0.960

The kink class achieves by far the lowest mean structural energy ($\bar{\mathcal{E}}_{\text{kink}} \approx 0.015$) and captures essentially all the Boltzmann weight ($\sum P_w \approx 99.96\%$). The strong weight concentration on kink configurations should be interpreted as a benchmark-level demonstration of discriminative power. Since the activity window and sector weights are still benchmark choices, the numerical dominance reported here does not yet establish model-independent universality.

The physical interpretation is transparent:

- **Vacuum** configurations have $s_{\text{dyn}} \approx 0.013$, far below the activity target $s_\star = 0.42$, yielding a large activity penalty.
- **Kink** configurations evolve with $s_{\text{dyn}} \approx 0.255$, close to the activity target, high spatial connectivity (large mutual information), and remain within the physical field range.
- **Localised** packets disperse rapidly, losing spatial coherence and accumulating kinetic energy.
- **Chaotic** configurations have $s_{\text{dyn}} \approx 0.960$, far above the activity target, and minimal spatial structure.

4.2 Robustness Across Parameter Sweep

To test whether kink dominance depends sensitively on the choice of λ_i and β, we perform a systematic sweep over:

$$\lambda_{\text{stab}}, \lambda_{\text{conn}}, \lambda_{\text{dyn}} \in \{0.5,\ 1.0,\ 2.0\}, \qquad \beta \in \{2.0,\ 4.0,\ 8.0\}. \tag{7}$$

This yields $3^3 \times 3 = 81$ distinct parameter combinations.

Table 2: Robustness of class dominance: number of parameter settings (out of 81) in which each class achieves the highest total Boltzmann weight.

Class	Wins (out of 81)
Vacuum	0
Kink	**81**
Localised	0
Chaotic	0

The result is unambiguous: *kink configurations dominate across all 81 parameter settings.* No other class wins under any combination of weights or inverse temperatures tested.

This robustness is the central result of this paper. It shows that the structural preference for kink configurations is not an artefact of a particular parameter choice, but a stable property of the selection functional across a wide region of the tested parameter space. We note that this demonstrates robustness within the benchmark setup, not universality across nonlinear field theories.

5 Discussion

5.1 Why Kinks Are Preferred

The structural selection of kink configurations can be understood from the interplay of the three diagnostic sectors:

Activity window: Kinks evolve with moderate dynamical activity ($s_{\text{dyn}} \approx 0.25$), close to the preferred window $s_\star = 0.42$. Vacuum states are too static; chaotic configurations are too active. Only kinks and localised packets occupy the intermediate regime, with kinks being more stable.

Connectivity: The spatial coherence of a kink — a smooth, extended transition between two vacuum regions — maximises the mutual information between adjacent field values. Homogeneous vacuum states and chaotic configurations have low spatial information content.

Stability: Kinks are topologically protected: they cannot decay to the vacuum without crossing an energy barrier. This translates directly into low kinetic energy fluctuations and no boundary violations, minimising the stability sector $\mathcal{E}_{\text{stab}}$.

Together, these three properties identify kinks as the class that best satisfies all structural criteria simultaneously — a convergence that no other class achieves.

5.2 Connection to Complex Systems Theory

A recurring observation in complex systems research is that persistent configurations tend to be locally stable, respect conservation channels, remain dynamically active, and preserve structural coherence [2]. The present results confirm this observation quantitatively: kink configurations score optimally on all structural criteria, while the other classes fail on at least one.

The result also suggests that the structural functional may not be entirely external to the field theory. The fact that it selects topologically protected configurations — whose stability follows from fundamental homotopy properties of the field space — points toward a connection between structural selection and the intrinsic geometry of the solution manifold, as Rajaraman [4] discusses in the context of soliton stability.

5.3 Relation to the Companion Framework

The present study extends the companion paper [1] in two directions:

1. It provides the first multi-class numerical test, going beyond the single-topology toy model of the companion paper.

2. It introduces a *preferred activity window* for $\mathcal{S}_C$, replacing the monotone activity reward with a symmetric penalty around a target activity level, which is more physically motivated and consistent with Prigogine's framework of regulated dissipative structures [2].

5.4 Limitations and Open Questions

The present study has several important limitations:

1. **One spatial dimension.** The result needs to be verified in 2+1 and 3+1 dimensions, where the topological landscape is richer and the competition between different soliton types more complex.

2. **Conservative dynamics.** The Klein–Gordon equation is conservative; the selection operates post-hoc over an ensemble. Extension to dissipative or driven systems would provide a more direct test of the selection mechanism.

3. **Mutual information estimator.** The connectivity sector uses a histogram-based mutual information estimator with 16 bins. This is a practical benchmark choice; alternative estimators and binning schemes should be tested in future work to verify robustness of the connectivity ranking.

4. **Parameter determination.** The Lagrange multipliers λ_i and activity target $s_\star$ are chosen as benchmark values. A first-principles derivation of these parameters from information-balance conditions remains open.

5. **Quantum extension.** The present framework is classical. The extension to quantum field theories, where the solution manifold is replaced by a space of quantum states, requires separate investigation.

6 Conclusion

We have presented a numerical study of structural selection in a one-dimensional ϕ^4 scalar field theory. Using the structural energy functional introduced in the companion paper [1], we evaluated four classes of field configurations — vacuum, kink, localised, and chaotic — and assigned Boltzmann weights based on their structural energy.

The central result is robust and unambiguous: kink configurations dominate the structural weight across all 81 combinations of Lagrange multiplier and inverse temperature values tested, with a weight of $\approx 99.96\%$ at the reference parameter setting.

This result is consistent with the core prediction of the structural selection framework: among all field configurations compatible with the dynamics, topologically stable, moderately active, and spatially coherent configurations are structurally preferred. The selection does not appear to be an artefact of a particular parameter choice, but rather a stable feature of the functional across the tested parameter space.

The framework is presented as a research programme. Future work will extend the analysis to higher dimensions, dissipative systems, multi-field theories, and the quantum setting.

A Reproducibility

All results are fully reproducible from the companion code:

https://github.com/Chris4081/structural-selection-principle

Key parameters:

$$\begin{aligned} N = 128, \quad \Delta x = 1, \quad \Delta t = 0.02, \quad T_{\max} = 24, \\ n = 12 \text{ runs per class}, \quad \text{seed} = 42, \\ s_\star = 0.42, \quad \sigma_s = 0.18, \quad \alpha = 0.5, \\ \phi_{\max} = 3, \quad \Lambda_{\text{stab}} = \Lambda_{\text{bound}} = 10. \end{aligned}$$

Run with:

```
pip install numpy pandas matplotlib
python maat_structural_selection_study_v2.py
```

References

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